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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

BACKLOG SUBJECT EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	:	**B.E :- Common to CSE (CY) / CSE (AI & ML) / (AI & ML) / (AI & DS)**	**Semester**	:	**III**
Course Name	:	**Discrete Mathematical Structures**	**Max. Marks**	:	**100**
Course Code	:	**CY35 / CI35 / AI34 / AD34**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

1.	a)	Prove by contra positive "If $3n+2$ is odd then n is odd."	CO1	(06)
	b)	Construct the truth table for the compound statement $\sim(p \vee \sim q) \rightarrow \sim p$, where $p, q, r,$ are primitive statements.	CO1	(07)
	c)	Prove by mathematical induction that, for all positive integer, 7 divides $3^{2n+1} + 2^{n+2}$.	CO1	(07)
2.	a)	Write the inverse and converse of "A positive integer is a prime only if it has no divisors other than 1 and itself."	CO1	(06)
	b)	Prove that for any three propositions p, q, r the compound proposition given by $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology.	CO1	(07)
	c)	Prove by mathematical induction that, for all positive integers $\mathrm{n} \geq 1$, $1+2+3+\cdots+\mathrm{n} = \frac{\mathrm{n(n+1)}}{2}$.	CO1	(07)

UNIT – II

3.	a)	If A={1,2,3,5,30} and R is the divisibility relation , prove that (A,R) is a lattice . Hence draw the Hasse diagram.	CO2	(06)
	b)	The relations R and S on the set A={a,b,c} are as follows : R={(a,a), (a,b), (a,c), (b,c)} S= {(a,c),(b,a), (b,c), (c,c)} Find $\overline{R}$, R^{-1} ,$R \cup S$, $R \cap S$, S^{-1}. Also draw their diagraphs and obtain matrix of each of the relations.	CO2	(07)
	c)	Let $A = \{1,2,3\}$ and $B = \{1,2,3,4,5\}$. The function $f: A \rightarrow B$ given by $f = \{(1,1), (2,3), (3,4)\}$. Test whether *f* is (i) one-one (ii) onto.	CO2	(07)
4.	a)	Check whether D_{20}, the set of all divisors of 20 is a Boolean algebra.	CO2	(06)
	b)	Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and test a relation R defined as $R = \{(x, y): \|x - y\| = 2\}$ for equivalence relation.	CO2	(07)
	c)	Let $A = \{1, 2, 3\}$ and B= $\{4, 6\}$ define the relation R such that "<" then find relation, complimentary and inverse relations.	CO2	(07)

UNIT – III

5.	a)	Find the number of ways in which 5 boys and 3 girls may be arranged in a row so that (i) all girls may be together (ii) no two girls may be together.	CO3	(06)
	b)	One hundred tickets, numbered 1, 2,….,100 are sold to 100 different peoples for a drawing. Four different prizes are awarded including a grand prize (a trip to Srilanka). How many ways are there to award prizes if, (i) No Restrictions (ii) the person holding ticket numbered 50 wins the grand prize (iii) the person holding ticket numbered 50 wins one of the prizes.	CO3	(07)
	c)	Solve $a_n=6a_{n-1}-9a_{n-2}$, $a_0=1$, $a_1=6$.	CO3	(07)

6. a) How many words can be formed from the letters of the word "MONDAY", if (i) all letters used at a time (ii) 4 letters are used at a time. CO3 (06)

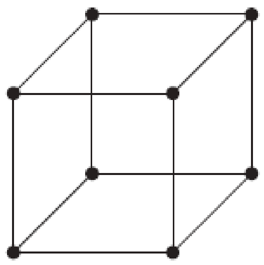
   b) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are selected. CO3 (07)

   c) Solve the recurrence relation: $a_n = 3a_{n-1} + 1,\ \ n \geq 1$; given $a_0 = 1$. CO3 (07)

UNIT – IV

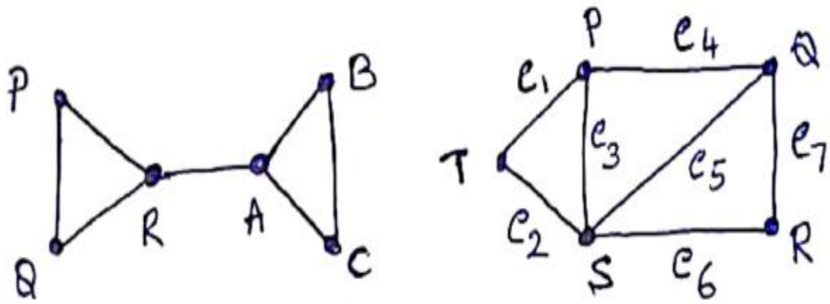
7. a) Define: CO4 (06)
   i) walk in a graph
   ii) path in a graph
   iii) Hamiltonian graph with an example for each.

   b) Define Planar graph and chromatic number. Determine whether the following graph is planar. If so, represent the graph. CO4 (07)

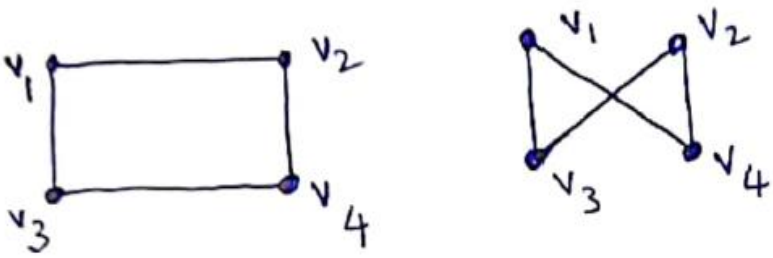


   c) Find the graph which has the incidence matrix $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$ and also write the corresponding adjacency matrix CO4 (07)

8. a) Explain which of the following graph, (i) contains an Euler trail (ii) not contains an Euler circuit. CO4 (06)



   b) Define isomorphic graphs and prove that the two graphs shown below are isomorphic. CO4 (07)



   c) Write the graph for which the adjacency matrix $\begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$. Also write the corresponding incidence matrix. CO4 (07)

UNIT-V

9.	a)	Define semigroup and monoid. Determine whether the set $(Z^+,*)$ is a semigroup and monoid.	CO5	(06)
	b)	State and prove Fermat's last theorem.	CO5	(07)
	c)	Let G be the set of non zero real numbers. Where * is defined by a*b=ab/5. Verify (G,*) is an abelian group.	CO5	(07)
10.	a)	Define Homomorphism and isomorphism of a group. Let (R^+, x) and (R,+) be two semigroups. Where R^+ is set of all positive real numbers , R is the set of all real numbers , x is usual multiplication and + is usual addition . Let f : $R^+ \rightarrow R$ defined by $f(x) = \log_e x \ \forall \ x \in R^+$. Verify that f is an isomorphic mapping.	CO5	(06)
	b)	Let G be the set of real numbers not equal to -1, and * be defined by a*b=a+b+ab. Verify that (G,*) is an abelian group.	CO5	(07)
	c)	Using Fermat's theorem, solve $24x \equiv 11(mod 7)$.	CO5	(07)

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